

Shilpa Sundar, PharmD, MPS, PhD (in progress)

PhD researcher specializing in graph-based deep learning and scalable ML systems, with experience in knowledge graphs and applied AI for biomedical decision-making

WORK EXPERIENCE

Translational and Integrative Sciences Lab (TISLab)

Aug 2024 – Present

Graduate Research Assistant

- Research integrates knowledge graphs, ontologies, and deep learning methods with a focus on genetic rare disease applications. Research includes NCATS Biomedical Data Translator, Every Cure Matrix drug repurposing for rare diseases, designed graph-based learning pipelines with iterative model refinement and evaluation loops (agent-like workflows), and developing scalable models for drug-drug interaction prediction models using graph-based deep learning methods for biomedical knowledge representation and extraction

Innovate Carolina: University of North Carolina at Chapel Hill

Dec 2023 – May 2024

Graduate Research Assistant

- Conducted comprehensive market, patent, and industry research for innovators and help entrepreneurs launch startups by analyzing data to support novel product commercialization

Nucleate Startup Accelerator Program: Research Triangle Park

Nov 2023 – May 2024

Nucleate Activator Finalist

- Lead business consultant in the 2023 Activator Cohort, a program that provides mentorship, educational workshops, and networking opportunities to the next generation of bioentrepreneurs

Burlington Medical Fabrics, an Elevate Textiles company

Sep 2023 – Dec 2023

Quality Improvement Consultant

- Led a quality improvement project by reducing error rates to below 5% in fabric processing utilizing Lean Six Sigma techniques (DMAIC, and 5Whys) and optimizing equipment use for high-value tasks

UNC Health Hospital: Department of Radiation Oncology

Aug 2023 – July 2024

Graduate Research Assistant

- Implement a comprehensive study on burnout in surgical teams, employing systems analysis and human factors interaction approach, contextual inquiries, focused-group discussions, and surveys to assess stress factors. Analyzed data using advanced tools, identifying key patterns, and providing recommendations to the leadership to enhance well-being and efficiency in surgical environments

Trinity Life Sciences (USA – based)

Dec 2022 – May 2023

Associate Consultant

- Led team in commercial analytics, analyzing primary and secondary market research data to deliver insights for pharma and MedTech clients. Responsibilities included data cleaning, analysis, insights generation, and visualization, working with datasets such as Symphony APLD claims, Breakaway formulary, PantherRx sales, Veeva Open Affiliations, UBC, IntrinsicIQ, and Texas Oncology data

Clarivate Analytics (USA – based)

Feb 2019 – Oct 2022

Associate Consultant ← Research Associate ← Intern

- End-to-end execution: Pricing and market access strategy consulting for EU4, UK, JAPAC, and US healthcare markets, providing go/no-go decisions, market expansion, forecasting, and business recommendations using primary and secondary research; developed a Social Determinants of Health database; conducted KOL, policymaker, and payer interviews; collaborated with pharma and PBM companies on market access strategy; trained and mentored interns and new hires

Kasturba Medical Hospital

Jan 2018 – Feb 2019

Hospital Pharmacist Intern

- Trained as hospital and clinical pharmacist in specific clinical departments, including general medicine, neurosurgery, neonatology, and neurology; trained in prescription management, drug dosage adjustment, clinical interventional plan, and medication counseling

AWARDS AND EXTRA-CURRICULAR

- Recent award:** Finalist at the Nucleate RTP chapter of the startup accelerator program – awarded in May 2024 for scientific excellence and diversity, equity, and inclusion; Winner of the 2023 Penn Healthcare case competition (A Wharton Business School Organization) - case study in innovation and leveraging AI/ML for value-generation in rare diseases
- Treasurer and Vice President at the Bangalore Toastmaster's Club:** Bangalore Chapter in 2022
- Social Service:** Mentor for under-served African students with Global Mentorship Initiative 2021

CERTIFICATIONS

Green belt in Lean Six Sigma | Tableau Proficiency Certificate | Agile Foundations | Healthcare Marketplace Specialization | Market Access Certification | Group 1: Biomedical IRB CITI Training

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LinkedIn: <https://www.linkedin.com/in/drshilpasundar/>

EDUCATION

University of North Carolina at Chapel Hill, USA

PhD candidate

Fall 2024 Intake (full-time: residential: STEM Designated)

- Concentration in Computational Knowledge Engineering, Health Informatics, and Genetics
- Research Interests: Clinical and health informatics, knowledge graphs, deep neural networks, predictive modeling, drug repurposing, ontologies, and data harmonization

University of North Carolina at Chapel Hill, USA

MPS in biomedical and health informatics; GPA: 4.0

2023-2024 (full-time: residential: fast-track: STEM Designated)

- Merit-based scholarship recipient
- Concentration in Clinical Informatics, Programming, and Visualization

Manipal University, India

Doctor of Pharmacy (PharmD); GPA: 3.8

2013-2019

- Distinction with top 5 rank holder through all 6 years
- Dissertation thesis: Assessment of risk factors in stroke and myocardial infarction in South India: A retrospective study

University of British Columbia, Canada

VSP MED – summer program

2017

- Awarded a scholarship and secured 92%

TOOLS AND DATABASE

SQL, BigQuery, Python (Kedro, NumPy, Pandas, PyTorch, DGL, TensorFlow, Scikit-Learn, StellarGraph), Neo4j (Cypher), Graph Data Science Library (GDS), Tableau, Power BI, SAS JMP, SPSS, OMOP CDM, LOINC, EPIC, SNOMED CT, ICD-10-CM, EHR systems, Cortellis, Kentico, GlobalData, LexisNexis, BrandWatch, Fingertip Formulary, Microsoft Office Suite, Figma

SKILLS

- Machine Learning & Deep Learning
- Graph Knowledge Engineering (Knowledge Graphs, Neo4j)
- Biomedical AI & Drug Interaction Prediction
- Clinical Data Science & Advanced Data Analytics
- Strategy, Market Access, & Management Consulting (Pricing, Reimbursement)

APPLIED ML PROJECTS & HACKATHONS

AWS Open Science Hackathon on Knowledge Graphs (2025, NVIDIA-sponsored)

- Built and analyzed large-scale scientific knowledge graphs
- Applied graph ML / embeddings for discovery tasks
- Collaborated in a fast-paced applied research environment